

Olympiad Test Paper — Mathematics (Class 7) — Paper 5

Chapter 12: Another Peek Beyond the Point

Paper 5 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	12 — Another Peek Beyond the Point
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. At this level the deadliest slips are losing a power of ten while chaining several multiplications and divisions, scaling only one number when dividing by a decimal, miscounting decimal places in a product, and confusing a place-value shift with appending zeros; estimate the order of magnitude of every answer before committing to a decimal point. Do not write on this paper — mark answers on your answer sheet.

1. Compute $(0.025 \times 0.4 \times 1000) \div 0.05$.

- (A) 20 (B) 2
(C) 2000 (D) 200

2. A shirt is marked at a price, then discounted by 0.25 of that price. Writing the discount as a fraction in lowest terms gives:

- (A) 25/100 (B) 1/40
(C) 1/4 (D) 2/5

3. A 1.9 m ribbon is cut into pieces 0.12 m long; whatever is left over is then cut into pieces 0.03 m long. Counting full pieces of BOTH sizes, how many pieces are produced in total?

- (A) 15 (B) 16
(C) 18 (D) 63

4. A number multiplied by 0.004 gives 0.12. The number is:

- (A) 3 (B) 30
(C) 300 (D) 0.3

5. How many decimal places does $(0.05)^2 \times (0.2)^3$ have, written out fully before any trailing zeros are dropped?

- (A) 5 (B) 6
(C) 8 (D) 7

6. The number 5.6 is multiplied by 1000 and the number 56000 is divided by 100. By how much does the first result exceed the second?

- (A) 504 (B) 50400
(C) 5.04 (D) 5040

7. Expressed as a percentage, the decimal 0.0375 equals:

- (A) 37.5% (B) 0.375%
(C) 3.75% (D) 0.0375%

8. Estimate first, then identify the exact value: $4.95 \times 19.8 \div 0.099$.

- (A) 99 (B) 990
(C) 9900 (D) 9.9

9. Compute $((2.4 \div 0.6) + (0.9 \div 0.3)) \times 0.05$.

- (A) 0.35 (B) 3.5
(C) 0.035 (D) 35

10. If 0.8 kg of cheese costs ₹420, the cost of 2.5 kg of the same cheese is:

- (A) ₹131.25 (B) ₹1050
(C) ₹1312.50 (D) ₹525

11. A water tank loses 0.015 litre per second through a leak. The litres lost in 1.2 hours are:

- (A) 6.48 (B) 64.8
(C) 648 (D) 1.08

12. When 0.0009 is multiplied by a power of ten the result is 9. The number of zeros after the 1 in that power of ten is:

- (A) 3 (B) 5
(C) 4 (D) 2

13. Compute $(1.2)^2 - (0.8)^2$.

- (A) 0.08 (B) 0.8
(C) 8 (D) 0.16

14. The number 3.0405 is multiplied by 1000. In the result, the digit in the tens place is:

- (A) 0 (B) 4
(C) 3 (D) 5

15. Three-eighths of a quantity is taken, written as a decimal and then as a percent. The percent is:

- (A) 3.75% (B) 375%
(C) 37.5% (D) 0.375%

16. Estimate first, then give the exact value: $0.0784 \div 0.0098$.

- (A) 0.8 (B) 8
(C) 80 (D) 0.08

17. Compute $(0.6 \times 0.5 \times 0.4) \div (0.2 \times 0.3)$.

- (A) 20 (B) 0.2
(C) 2 (D) 0.02

18. A car covers 0.6 km on every 0.04 litre of fuel. Its mileage, in km per litre, is:

- (A) 15 (B) 1.5
(C) 150 (D) 0.15

19. Compute $7.2 \div (0.3 \times 0.4) - 10$.

- (A) 5 (B) 50
(C) 60 (D) 0.5

20. By what number must 0.0125 be DIVIDED to give 0.5?

- (A) 0.25 (B) 0.0025
(C) 2.5 (D) 0.025

21. Compute $(0.04)^2 + (0.3)^2$.

- (A) 0.916 (B) 0.00916
(C) 0.0916 (D) 0.34

22. In 47.0813, by how much does the value increase if the digit 8 is replaced by 9?

- (A) 0.01 (B) 0.1
(C) 0.001 (D) 1

23. A debt is $\frac{1}{8}$ of a salary. Written as a percentage of the salary, the debt is:

- (A) 1.25% (B) 8%
(C) 12.8% (D) 12.5%

24. Estimate first, then give the exact value: $0.0306 \div 0.017$.

- (A) 18 (B) 1.8
(C) 0.18 (D) 180

25. Compute $(2.5 \times 0.04) \div (0.5 \times 0.2) + 1$.

- (A) 3 (B) 11
(C) 2 (D) 1.1

26. Rice costs ₹62.50 per kg. A shopper buys 2.4 kg, pays with two ₹100 notes, and receives change. The change is:

(A) ₹150

(B) ₹15

(C) ₹50

(D) ₹35

27. A length of cable weighs 0.024 kg per metre. A coil weighs 1.44 kg. Half the coil is sold; the metres remaining are:

(A) 30

(B) 60

(C) 3

(D) 300

28. Compute $(0.9 \div 0.06) \times (0.08 \div 0.4)$.

(A) 3

(B) 30

(C) 0.3

(D) 0.03

29. How many decimal places does $(0.12)^2$ have when written out fully before trimming?

(A) 2

(B) 4

(C) 3

(D) 6

30. The number 6.05 is multiplied by 100, then 40 is subtracted. The digit in the tens place of the result is:

(A) 5

(B) 0

(C) 6

(D) 4

31. A coupon gives 0.2 off the price as a fraction of the price. Written as a percent discount, this is:

(A) 2%

(B) 0.2%

(C) 200%

(D) 20%

32. A plank 4.5 m long has pieces of 0.85 m, 1.15 m and 0.6 m cut off; the rest is divided into 0.4 m strips. The number of full 0.4 m strips is:

(A) 5

(B) 4

(C) 3

(D) 47

33. Compute $(1.5 + 2.5) \div 0.016$.

(A) 250

(B) 25

(C) 2500

(D) 2.5

34. If 1.6 metres of cloth cost ₹120, the cost of 2.5 metres at the same rate is:

(A) ₹18.75

(B) ₹187.50

(C) ₹300

(D) ₹75

35. Estimate first, then give the exact value: 198×0.0049 .

(A) 9.702

(B) 0.09702

(C) 97.02

(D) 0.9702

36. By what number must 0.06 be multiplied to give the same result as $0.06 \div 0.0025$?

- (A) 400 (B) 40
(C) 4000 (D) 4

37. How many decimal places does the product 0.025×0.0004 have before trailing zeros are dropped?

- (A) 6 (B) 8
(C) 7 (D) 5

38. A man earns ₹0.45 per item and makes 1.2 thousand (= 1200) items, then spends $\frac{1}{3}$ of the total. The amount he keeps is:

- (A) ₹540 (B) ₹180
(C) ₹36 (D) ₹360

39. The number 0.7 is multiplied by 1000, then divided by 100, then multiplied by 0.5. The result is:

- (A) 3.5 (B) 35
(C) 0.35 (D) 0.035

40. Written as a decimal, the percentage 0.6% equals:

- (A) 0.06 (B) 0.006
(C) 0.6 (D) 6

41. Estimate first: the value of $0.0504 \div 0.012$ is closest to which whole number?

- (A) 4 (B) 40
(C) 5 (D) 0.4

42. Compute $(0.36 \div 0.6) + (0.36 \div 0.06)$.

- (A) 66 (B) 0.66
(C) 6.06 (D) 6.6

43. A jug holds 0.35 litre. From a tank of 9.8 litres, the number of full jugs that can be filled, with the leftover, is:

- (A) 27 jugs, 0.35 litre left (B) 28 jugs, 0 litres left
(C) 28 jugs, 0.15 litre left (D) 2.8 jugs

44. Compute $((0.2)^3 \times 1000) \div 0.4$.

- (A) 2 (B) 200
(C) 0.2 (D) 20

45. Which fraction in lowest terms equals the percent 37.5%?

- (A) $\frac{3}{8}$ (B) $\frac{375}{1000}$
(C) $\frac{3}{80}$ (D) $\frac{37}{100}$

46. Estimate first, then give the exact value: $50.4 \div 0.0072$.

- (A) 7000 (B) 700
(C) 70000 (D) 70

47. Compute $(4.8 \div 0.6) \div (0.16 \div 0.02)$.

- (A) 10 (B) 0.1
(C) 100 (D) 1

48. A worker's hourly wage of ₹0.025 thousand (= ₹25) is paid for 7.5 hours; she then donates 0.2 of the pay. The amount she keeps is:

- (A) ₹187.50 (B) ₹37.50
(C) ₹15 (D) ₹150

49. If 0.6 of a number is 0.45, then 0.4 of the same number is:

- (A) 3 (B) 0.3
(C) 0.03 (D) 0.18

50. Compute $(0.05)^2 \div (0.5 \times 0.005)$.

- (A) 10 (B) 0.1
(C) 1 (D) 100

51. Compute $0.0144 \div 0.012 + 0.0144 \div 0.0012$.

- (A) 132 (B) 1.32
(C) 13.2 (D) 13.02

52. The decimal 1.04 is written as a percentage. It equals:

- (A) 104% (B) 1.04%
(C) 10.4% (D) 1040%

53. In 8.2059, the digit in the thousandths place is multiplied by 10 and added to the digit in the tenths place. The result is:

- (A) 7 (B) 25
(C) 50 (D) 52

54. Written as a decimal in lowest form, the value of 0.875 expressed back from its fraction $7/8$ confirms which percent?

- (A) 8.75% (B) 0.875%
(C) 87.5% (D) 875%

55. Estimate first: $0.0399 \div 0.0021$ is closest to:

- (A) 1.9 (B) 190
(C) 2 (D) 19

56. Compute $(0.8 \times 0.75 \times 0.5) \div 0.3$.

- (A) 10 (B) 1
(C) 0.1 (D) 0.01

57. A bottle of medicine holds 0.18 litre. Each dose is 0.015 litre. After all full doses are poured, the leftover is:

- (A) 0 litre (B) 0.015 litre
(C) 0.003 litre (D) 0.18 litre

58. Place in increasing order: $P = 0.6 \times 0.5$, $Q = 0.6 \div 0.5$, $R = 0.6 + 0.5$, $S = 0.6 - 0.5$.
The correct order is:

- (A) $S < P < R < Q$ (B) $S < P < Q < R$
(C) $P < S < Q < R$ (D) $S < Q < P < R$

59. A 0.25 kg packet of saffron costs ₹400. A buyer wants 0.6 kg. The cost is:

- (A) ₹960 (B) ₹96
(C) ₹1600 (D) ₹240

60. A roll of cloth 5.8 m long is cut into 0.8 m pieces. The number of full pieces and the leftover length are:

- (A) 7 pieces, 0 m left (B) 6 pieces, 0.2 m left
(C) 8 pieces, 0.2 m left (D) 7 pieces, 0.2 m left

Solutions — Mathematics (Class 7), Chapter 12: Another Peek Beyond the Point — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	B	21	C	31	D	41	A	51	C
2	C	12	C	22	A	32	B	42	D	52	A
3	C	13	B	23	D	33	A	43	B	53	D
4	B	14	B	24	B	34	B	44	D	54	C
5	D	15	C	25	C	35	D	45	A	55	D
6	D	16	B	26	C	36	A	46	A	56	B
7	C	17	C	27	A	37	C	47	D	57	A
8	B	18	A	28	A	38	D	48	D	58	A
9	A	19	B	29	B	39	A	49	B	59	A
10	C	20	D	30	C	40	B	50	C	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Work left to right keeping track of size. $0.025 \times 0.4 = 0.01$ ($25 \times 4 = 100$, with $3 + 1 = 4$ decimal places gives 0.0100). Then 0.01×1000 shifts the point 3 right = 10. Finally $10 \div 0.05$: scale both by 100 to get $1000 \div 5 = 200$. Dropping one of the powers of ten while chaining lands on 20 or 2000.

2. (C) $0.25 = 25/100$; dividing numerator and denominator by 25 gives $1/4$. Leaving it as $25/100$ is unsimplified; $1/40$ would be 0.025 , ten times too small; $2/5 = 0.4$.

3. (C) First cut: $1.9 \div 0.12$, scale both by 100 to get $190 \div 12 = 15$ full pieces using $15 \times 0.12 = 1.80$ m, leaving $1.9 - 1.80 = 0.10$ m. Second cut: $0.10 \div 0.03$, scale by 100 to get $10 \div 3 = 3$ full pieces (using 0.09 m), leaving 0.01 m. Total full pieces = $15 + 3 = 18$. Stopping after the first cut gives 15; mis-counting the leftover gives 16; treating every piece as a 0.03 m piece gives 63.

4. (B) Divide $0.12 \div 0.004$. Scale both by 1000: $120 \div 4 = 30$. Dividing $0.12 \div 4 = 0.03$ forgets to scale the divisor; an extra power of ten gives 300 or 3.

5. (D) $(0.05)^2$ needs $2 + 2 = 4$ places, $(0.2)^3$ needs $1 + 1 + 1 = 3$ places; a product of the two carries $4 + 3 = 7$ decimal places. (The digit product is $25 \times 8 = 200$, and $0.0025 \times 0.008 = 0.0000200$, which indeed sits at the seventh place before trimming.) Adding instead of summing place counts, or stopping at the larger of the two, gives 4 or 3-based wrong totals.

6. (D) $5.6 \times 1000 = 5600$ (shift 3 right). $56000 \div 100 = 560$ (shift 2 left). Difference $5600 - 560 = 5040$. Subtracting before shifting, or one wrong shift, gives 504 or 50400.

7. (C) To turn a decimal into a percent, multiply by 100 (shift the point two places right): $0.0375 \times 100 = 3.75$, so 3.75%. Shifting only one place gives 0.375%; shifting three gives 37.5%; forgetting to shift gives 0.0375%.

8. (B) Estimate $5 \times 20 \div 0.1 = 100 \div 0.1 = 1000$, so the answer is near 1000. Exactly: $4.95 \times 19.8 = 98.01$; then $98.01 \div 0.099$, scale both by 1000 to get $98010 \div 99 = 990$. Misplacing the point gives 99 or 9900.

9. (A) Inside: $2.4 \div 0.6 = 4$ ($24 \div 6$) and $0.9 \div 0.3 = 3$ ($9 \div 3$); their sum is 7. Then $7 \times 0.05 = 0.35$. Doing 7×0.5 by misreading the factor gives 3.5; misplacing the point gives 0.035 or 35.

10. (C) Unit rate = $420 \div 0.8 = 525$ per kg (scale both by 10: $4200 \div 8 = 525$). Then $525 \times 2.5 = 1312.50$. Stopping at the per-kg rate gives 525; multiplying 420×2.5 forgets to find the rate first and gives 1050; misplacing the point gives 131.25.

11. (B) 1.2 hours = $1.2 \times 3600 = 4320$ seconds. Loss = $0.015 \times 4320 = 64.8$ litres ($15 \times 4320 = 64800$, with three decimal places $\rightarrow 64.800$). Using minutes (72) instead of seconds gives 1.08; misplacing the point gives 6.48 or 648.

12. (C) Moving the point from 0.0009 to 9 is 4 places right, which is $\times 10000$ — a 1 followed by 4 zeros. Counting only 3 places ($\times 1000$) gives 0.9, not 9.

13. (B) $(1.2)^2 = 1.44$ and $(0.8)^2 = 0.64$. Difference $1.44 - 0.64 = 0.80 = 0.8$. (Checking by the identity $(1.2 - 0.8)(1.2 + 0.8) = 0.4 \times 2.0 = 0.8$ confirms it.) Squaring the difference 0.4 to get 0.16 is the classic trap; misplacing the point gives 0.08 or 8.

14. (B) 3.0405×1000 shifts the point 3 places right = 3040.5. Reading place values: 3 thousands, 0 hundreds, 4 tens, 0 ones, 5 tenths. So the tens digit is 4. The ones digit is 0, the thousands digit is 3, and the tenths digit is 5.

15. (C) $3/8 = 0.375$ (8 into 30 is 3 r6 $\rightarrow .3$; into 60 is 7 r4 $\rightarrow .07$; into 40 is 5 $\rightarrow .005$). As a percent, $0.375 \times 100 = 37.5\%$. Forgetting to multiply by 100 gives 0.375%; one too few shifts gives 3.75%.

- 16. (B)** Round to $0.08 \div 0.01 = 8$. Exactly: scale both by 10000, $784 \div 98 = 8$. Misplacing the point gives 0.8 or 80.
- 17. (C)** Numerator $0.6 \times 0.5 \times 0.4 = 0.12$ (digits $6 \times 5 \times 4 = 120$, three decimal places $\rightarrow 0.120$). Denominator $0.2 \times 0.3 = 0.06$. Then $0.12 \div 0.06$: scale both by 100, $12 \div 6 = 2$. Dividing $0.12 \div 6 = 0.02$ forgets to scale; an extra power of ten gives 20.
- 18. (A)** Mileage = $0.6 \div 0.04$: scale both by 100, $60 \div 4 = 15$ km per litre. Dividing $0.6 \div 4 = 0.15$ forgets to scale the divisor; an extra power of ten gives 150.
- 19. (B)** $0.3 \times 0.4 = 0.12$, then $7.2 \div 0.12$: scale both by 100, $720 \div 12 = 60$. Finally $60 - 10 = 50$. Subtracting first, or mishandling the division to 6, gives 5 or other wrong values.
- 20. (D)** If $0.0125 \div x = 0.5$, then $x = 0.0125 \div 0.5 = 0.025$ (scale both by 10: $0.125 \div 5 = 0.025$). A common error multiplies instead, giving 0.00625, or misplaces the point to 0.25 or 0.0025.
- 21. (C)** $(0.04)^2 = 0.0016$ ($4 \times 4 = 16$, four decimal places). $(0.3)^2 = 0.09$ (two decimal places). Sum = $0.0016 + 0.0900 = 0.0916$. Adding the bases ($0.04 + 0.3 = 0.34$) instead of the squares is the trap; misplacing the point gives 0.916 or 0.00916.
- 22. (A)** After the point the order is tenths(0), hundredths(8), thousandths(1), ten-thousandths(3). The 8 is in the hundredths place worth 0.08; replacing it with 9 makes it 0.09, an increase of 0.01. Reading it as tenths gives 0.1; as thousandths gives 0.001.
- 23. (D)** $1/8 = 0.125$, and $0.125 \times 100 = 12.5\%$. Forgetting one shift gives 1.25%; reading the denominator as the percent gives 8%; 12.8% is a miscalculation.
- 24. (B)** Estimate $0.03 \div 0.017 \approx 1.8$. Exactly: scale both by 1000, $30.6 \div 17 = 1.8$ (or $306 \div 170 = 1.8$). Misplacing the point gives 18 or 0.18.
- 25. (C)** Numerator $2.5 \times 0.04 = 0.1$ ($25 \times 4 = 100$, three decimal places $\rightarrow 0.100$). Denominator $0.5 \times 0.2 = 0.1$. Then $0.1 \div 0.1 = 1$, and $1 + 1 = 2$. Forgetting the +1 gives 1; mishandling the division gives 11 or 1.1.
- 26. (C)** Cost = $62.50 \times 2.4 = 150.00$ ($625 \times 24 = 15000$, three decimal places $\rightarrow 150.000$). Paid 200, change = $200 - 150 = 50$. Confusing cost with change gives 150; misplacing the point gives 15.
- 27. (A)** Full length = $1.44 \div 0.024$: scale both by 1000, $1440 \div 24 = 60$ m. Half remains, so $60 \div 2 = 30$ m. Forgetting to halve gives 60; mishandling the division gives 3 or 300.
- 28. (A)** $0.9 \div 0.06 = 15$ (scale by 100: $90 \div 6 = 15$). $0.08 \div 0.4 = 0.2$ (scale by 10: $0.8 \div 4 = 0.2$). Product $15 \times 0.2 = 3$. Losing a power of ten in either division gives 30, 0.3 or 0.03.

- 29. (B)** Each factor 0.12 has 2 decimal places, so the square has $2 + 2 = 4$ places: $12 \times 12 = 144 \rightarrow 0.0144$. Counting only 2 or 3 places misplaces the point.
- 30. (C)** $6.05 \times 100 = 605$ (shift 2 right). $605 - 40 = 565$. The tens digit of 565 is 6 (5 hundreds, 6 tens, 5 ones). Reading the ones digit gives 5; forgetting the subtraction leaves 605 whose tens digit is 0.
- 31. (D)** $0.2 \times 100 = 20$, so a 20% discount. Forgetting to multiply by 100 gives 0.2%; one too few shifts gives 2%; one too many gives 200%.
- 32. (B)** Cut total = $0.85 + 1.15 + 0.6 = 2.6$ m. Remaining = $4.5 - 2.6 = 1.9$ m. Strips = $1.9 \div 0.4 = 4.75$, so 4 full strips. Rounding up gives 5; mis-subtracting gives 3; treating $1.9 \div 0.4$ as $1.9 \div 4$ then scaling wrongly gives 47.
- 33. (A)** $1.5 + 2.5 = 4.0$, then $4 \div 0.016$: scale both by 1000, $4000 \div 16 = 250$. Dividing $4 \div 16 = 0.25$ forgets to scale the divisor; an extra power of ten gives 2500 or 25.
- 34. (B)** Unit rate = $120 \div 1.6 = 75$ per metre (scale by 10: $1200 \div 16 = 75$). Then $75 \times 2.5 = 187.50$. Stopping at the rate gives 75; multiplying 120×2.5 without finding the rate gives 300; misplacing the point gives 18.75.
- 35. (D)** Estimate $200 \times 0.005 = 1$, so the answer is near 1. Exactly: $198 \times 49 = 9702$, with $0 + 4 = 4$ decimal places $\rightarrow 0.9702$. Miscounting the place gives 9.702 or 0.09702.
- 36. (A)** $0.06 \div 0.0025$: scale both by 10000, $600 \div 25 = 24$. We need $0.06 \times x = 24$, so $x = 24 \div 0.06 = 400$ (scale by 100: $2400 \div 6 = 400$). Losing a power of ten gives 40 or 4000.
- 37. (C)** 0.025 has 3 decimal places and 0.0004 has 4, so the product has $3 + 4 = 7$ places: $25 \times 4 = 100 \rightarrow 0.0000100$, the 1 sitting at the fifth place but the full written form carries 7 places. Counting only the larger factor gives 4; adding wrongly gives 6 or 8.
- 38. (D)** Total = $0.45 \times 1200 = 540$ ($45 \times 1200 = 54000$, two decimal places $\rightarrow 540.00$). Spends $1/3 \rightarrow 540 \div 3 = 180$ spent, so keeps $540 - 180 = 360$ (which is $2/3$ of 540). Reporting the amount spent gives 180; reporting the total gives 540; misplacing the point gives 36.
- 39. (A)** $0.7 \times 1000 = 700$; $\div 100 = 7$; $\times 0.5 = 3.5$. Dropping the final factor gives 7; misplacing the point gives 35 or 0.35.
- 40. (B)** A percent means 'per hundred', so $0.6\% = 0.6 \div 100 = 0.006$ (shift the point two places left). Shifting only one place gives 0.06; forgetting to divide gives 0.6.
- 41. (A)** Round to $0.05 \div 0.012 \approx 4$. Exactly: scale both by 1000, $50.4 \div 12 = 4.2$ (or $504 \div 120 = 4.2$), closest to 4. Misplacing the point gives 40 or 0.4.

- 42. (D)** $0.36 \div 0.6 = 0.6$ (scale by 10: $3.6 \div 6 = 0.6$). $0.36 \div 0.06 = 6$ (scale by 100: $36 \div 6 = 6$). Sum $0.6 + 6 = 6.6$. Mis-aligning the place values gives 66, 0.66 or 6.06.
- 43. (B)** $9.8 \div 0.35$: scale both by 100, $980 \div 35 = 28$ exactly, so 28 full jugs and 0 litres left ($28 \times 0.35 = 9.80$). Stopping at 27 leaves a whole jug's worth; treating it as a non-integer gives 2.8.
- 44. (D)** $(0.2)^3 = 0.008$ (three decimal places). $\times 1000$ shifts 3 right = 8. Then $8 \div 0.4 = 20$ (scale by 10: $80 \div 4 = 20$). Stopping at $8 \div 0.4$ mishandled gives 2; an extra power of ten gives 200.
- 45. (A)** $37.5\% = 0.375 = 375/1000$; dividing numerator and denominator by 125 gives $3/8$. Leaving it as $375/1000$ is unsimplified; $3/80 = 0.0375$ (ten times too small); $37/100 = 0.37$.
- 46. (A)** Estimate $50 \div 0.007 \approx 7000$. Exactly: scale both by 10000, $504000 \div 72 = 7000$. Losing a power of ten gives 700 or 70000.
- 47. (D)** $4.8 \div 0.6 = 8$ ($48 \div 6$). $0.16 \div 0.02 = 8$ (scale by 100: $16 \div 2 = 8$). Then $8 \div 8 = 1$. Mishandling a place shift gives 10 or 0.1.
- 48. (D)** Pay = $25 \times 7.5 = 187.50$. Donates $0.2 \times 187.50 = 37.50$, keeps $187.50 - 37.50 = 150$ (which is 0.8×187.50). Reporting the donation gives 37.50; reporting the full pay gives 187.50; misplacing the point gives 15.
- 49. (B)** The number = $0.45 \div 0.6 = 0.75$ (scale by 10: $4.5 \div 6 = 0.75$). Then $0.4 \times 0.75 = 0.30 = 0.3$ ($4 \times 75 = 300$, three decimal places $\rightarrow 0.300$). Multiplying 0.4×0.45 directly gives 0.18; misplacing the point gives 3 or 0.03.
- 50. (C)** $(0.05)^2 = 0.0025$ (four decimal places). Denominator $0.5 \times 0.005 = 0.0025$ ($25 \times \dots$ gives $5 \times 5 = 25$ with $1 + 3 = 4$ places $\rightarrow 0.0025$). Then $0.0025 \div 0.0025 = 1$. Any place-shift slip gives 10 or 0.1.
- 51. (C)** $0.0144 \div 0.012 = 1.2$ (scale by 10000: $144 \div 120 = 1.2$). $0.0144 \div 0.0012 = 12$ (scale by 10000: $144 \div 12 = 12$). Sum $1.2 + 12 = 13.2$. Mis-aligning place values gives 132, 1.32 or 13.02.
- 52. (A)** Multiply by 100: $1.04 \times 100 = 104$, so 104%. Forgetting to shift gives 1.04%; one too few shifts gives 10.4%; one too many gives 1040%.
- 53. (D)** After the point: tenths = 2, hundredths = 0, thousandths = 5, ten-thousandths = 9. Thousandths digit $5 \times 10 = 50$, plus tenths digit $2 = 52$. Confusing the places gives 7 (2+5) or 25 (5 then 2).
- 54. (C)** $7/8 = 0.875$, and $0.875 \times 100 = 87.5\%$. Forgetting one shift gives 8.75%; forgetting both gives 0.875%; one too many gives 875%.

55. (D) Round to $0.04 \div 0.002 = 20$, so near 19. Exactly: scale both by 10000, $399 \div 21 = 19$. Misplacing the point gives 1.9 or 190.

56. (B) $0.8 \times 0.75 = 0.6$ ($8 \times 75 = 600$, three decimal places $\rightarrow 0.600$), then $\times 0.5 = 0.3$. Finally $0.3 \div 0.3 = 1$. Losing a power of ten in the chain gives 10 or 0.1.

57. (A) $0.18 \div 0.015$: scale both by 1000, $180 \div 15 = 12$ exactly, so 12 full doses and 0 leftover ($12 \times 0.015 = 0.180$). A miscount leaves 0.015 or a partial 0.003 litre.

58. (A) Compute each value: $P = 0.6 \times 0.5 = 0.30$; $Q = 0.6 \div 0.5 = 1.2$ (scale by 10: $6 \div 5 = 1.2$); $R = 0.6 + 0.5 = 1.1$; $S = 0.6 - 0.5 = 0.1$. In increasing order: $S (0.1) < P (0.3) < R (1.1) < Q (1.2)$. The trap is assuming division gives a small number and addition the largest, which swaps R and Q.

59. (A) Rate = $400 \div 0.25 = 1600$ per kg (scale by 100: $40000 \div 25 = 1600$). Then $1600 \times 0.6 = 960$. Stopping at the rate gives 1600; multiplying 400×0.6 without finding the rate gives 240; misplacing the point gives 96.

60. (D) $5.8 \div 0.8$: scale both by 10, $58 \div 8 = 7$ remainder 2, i.e. 7 full pieces using $7 \times 0.8 = 5.6$ m, leaving $5.8 - 5.6 = 0.2$ m. Claiming no leftover ignores the remainder; rounding the quotient up gives 8; under-counting gives 6.